Perimeter and Area

Test 3

Date: _____ Duration: 1 hour Maximum marks: 30

Use $\pi = \dfrac{22}{7}$. Draw a rough diagram wherever it helps. Show all your steps.

Paper design. The paper follows the CBSE pattern. Section A: 6×1 mark (4 multiple-choice and 2 assertion–reason questions). Section B: 3×2 marks. Section C: 3×3 marks. Section D: 1×5 marks. Section E: 1 case-based question of 4 marks. Total: 30 marks.

Competency focus. Question 9 tests direct application of a formula. The other questions are competency-based. Students compare figures, analyse an error, work backwards from a result, test a claim, or apply results to a real situation.

Concepts assessed. Area and circumference of a circle (Q1, 5, 7, 8, 13, 14); area and perimeter of a sector (Q2, 4, 9, 14); area of a segment (Q11); perimeter and area of a composite figure (Q12); triangles and parallelograms on the same base and between the same parallels (Q3); the median of a triangle (Q6, 10); Heron's formula (Q10).

Common misconceptions. Using the diameter in place of the radius (Q4, 7); leaving out the straight edges from the perimeter of a semicircle or sector (Q4, 12); assuming that figures of equal perimeter have equal areas (Q1); assuming that the area of a ring doubles when its width doubles (Q13).

Marking. Award method marks for a correct set-up, even where the final computation contains an arithmetic error.

Section A (1 mark each)

Questions 1 to 4 are multiple choice. Questions 5 and 6 are assertion–reason.

1. A circular rangoli and a square rangoli each have a perimeter of 88 cm. Which statement is true?

(a) The circle has the larger area

(b) The square has the larger area

(c) The two areas are equal

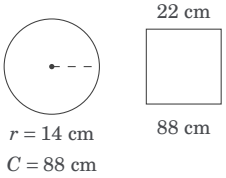
(d) The areas cannot be compared without more information

Answer.

(a).

Circle: $2 \times \dfrac{22}{7} \times r = 88$, so $r = 14$ cm. Area $= \dfrac{22}{7} \times 14 \times 14 = 616$ cm^2.

Square: side $= 88 \div 4 = 22$ cm. Area $= 22 \times 22 = 484$ cm^2.



2. A sector of a circle has radius 14 cm. The length of its arc is 22 cm. What is the area of the sector?

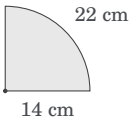
(a) 77 cm^2 (b) 154 cm^2 (c) 308 cm^2 (d) 616 cm^2

Answer.

(b).

Circumference $= 2 \times \frac{22}{7} \times 14 = 88$ cm. The arc is $\frac{22}{88} = \frac{1}{4}$ of the circumference.

So the sector is $\frac{1}{4}$ of the circle. Area $= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14 = 154$ cm^2.



3. A triangle and a parallelogram stand on the same base and lie between the same parallel lines. The area of the parallelogram is 48 cm^2. What is the area of the triangle?

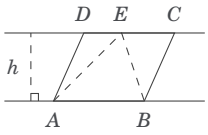
(a) 12 cm^2 (b) 24 cm^2 (c) 48 cm^2 (d) 96 cm^2

Answer.

(b).

The triangle and the parallelogram have the same base and the same height.

Area of the triangle $= \frac{1}{2} \times$ area of the parallelogram $= \frac{1}{2} \times 48 = 24$ cm^2.



4. A protractor is a semicircle of radius 7 cm. What is the perimeter of the protractor?

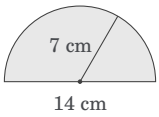
(a) 22 cm (b) 29 cm (c) 36 cm (d) 44 cm

Answer.

(c).

Curved edge $= \frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm. Straight edge = diameter = 14 cm.

Perimeter $= 22 + 14 = 36$ cm.



In Questions 5 and 6, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

5. Assertion (A): A circle with circumference 44 cm has an area of 154 cm^2.

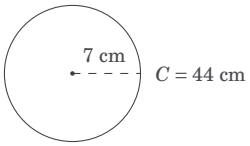
Reason (R): The area of a circle = $\frac{1}{2}\times$ circumference $\times$ radius.

Answer.

(a).

$2 \times \frac{22}{7} \times r = 44$, so $r = 7$ cm. Area $= \frac{1}{2} \times 44 \times 7 = 154$ cm^2. A is true.

R is true, because $\pi r^2 = \frac{1}{2} \times (2\pi r) \times r$. R gives the method used to obtain A, so R correctly explains A.



6. Assertion (A): A median of a triangle divides it into two triangles with equal perimeters.

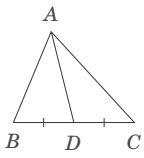
Reason (R): A median of a triangle divides it into two triangles with equal areas.

Answer.

(d).

A is false. The two triangles share the median and have equal halves of the base, but their third sides can differ in length.

R is true. The two triangles have equal bases and the same height.



Section B (2 marks each)

7. A circular cover for a well has a diameter of 14 cm. Ravi finds its area like this:

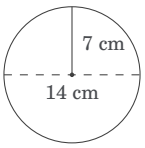
$$\text{Area} = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2.$$

Explain Ravi's mistake. Find the correct area of the cover.

Answer.

Ravi used the diameter, 14 cm, in place of the radius.

Radius $= 14 \div 2 = 7$ cm. Area $= \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$.

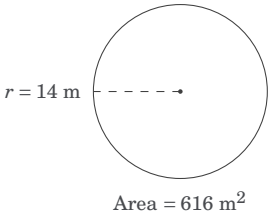


8. The area of a circular park is 616 m^2. Find the length of wire needed to fence the park once.

Answer.

$\frac{22}{7} \times r^2 = 616$, so $r^2 = 616 \times \frac{7}{22} = 196$ and $r = 14$ m.

Length of wire = circumference $= 2 \times \frac{22}{7} \times 14 = 88$ m.

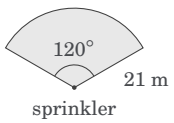


9. A lawn sprinkler sprays water up to a distance of 21 m. It turns through an angle of 120° and back. Draw a rough diagram. Find the area of the lawn that gets water.

Answer.

The watered part is a sector of radius 21 m and angle 120°.

Area $= \frac{120}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{3} \times 1386 = 462 \text{ m}^2$.



Section C (3 marks each)

10. A triangular plot of land has sides 26 m, 28 m and 30 m.

(a) Find the area of the plot using Heron's formula.

(b) The owner wants to divide the plot into two parts of equal area with one straight fence from a corner. Where should the fence meet the opposite side? Give a reason, and find the area of each part.

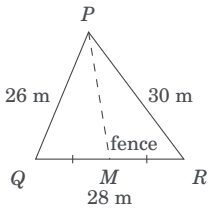
Answer.

(a) $s = \dfrac{26 + 28 + 30}{2} = 42$. $s - a = 16$, $s - b = 14$, $s - c = 12$.

Area $= \sqrt{42 \times 16 \times 14 \times 12} = \sqrt{112896} = 336$ m^2.

(b) The fence must meet the opposite side at its midpoint, so the fence is a median.

A median divides a triangle into two triangles of equal area. Each part $= 336 \div 2 = 168$ m^2.



11. In a circle with centre O and radius 14 cm, a chord AB makes an angle of 90° at the centre.

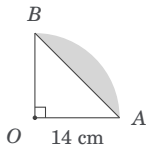
(a) Draw a rough diagram. Mark O, A, B, the radius and the angle. Shade the minor segment.

(b) Find the area of the minor segment.

(c) Find the area of the major segment.

Answer.

(a) See the diagram.



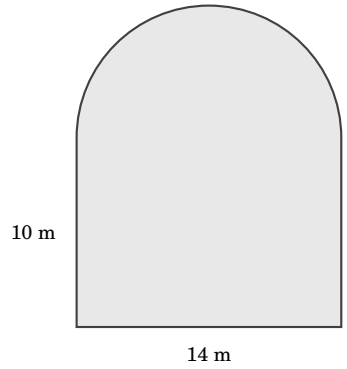
(b) Area of sector $OAB = \dfrac{90}{360} \times \dfrac{22}{7} \times 14 \times 14 = 154$ cm^2.

Area of triangle $OAB = \dfrac{1}{2} \times 14 \times 14 = 98$ cm^2, since $\angle AOB = 90°$.

Minor segment $= 154 - 98 = 56$ cm^2.

(c) Area of the circle $= \dfrac{22}{7} \times 14 \times 14 = 616$ cm^2. Major segment $= 616 - 56 = 560$ cm^2.

12. A school stage is a rectangle 14 m long and 10 m wide, with a semicircle on one of the 14 m sides, as shown.



(a) A rope is fixed along the whole edge of the stage. Find the length of the rope.

(b) Find the area of the stage floor.

Answer.

(a) The semicircle has diameter 14 m, so its radius is 7 m.

Curved edge $= \frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ m. Straight edges $= 10 + 14 + 10 = 34$ m.

Rope $= 22 + 34 = 56$ m.

(b) Rectangle $= 14 \times 10 = 140$ m^2. Semicircle $= \frac{1}{2} \times \frac{22}{7} \times 7 \times 7 = 77$ m^2.

Area $= 140 + 77 = 217$ m^2.

Section D (5 marks)

13. A circular pond in a temple garden has a radius of 14 m. A path 7 m wide runs around the pond, outside it.

(a) Draw a rough diagram. Find the area of the path. [2]

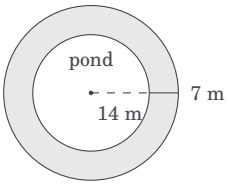
(b) The path is tiled at ₹ 40 per m^2. Find the cost of tiling. [1]

(c) Meena says, "If the path were 14 m wide, its area would be double." Is Meena right? Show your working and explain. [2]

Answer.

(a) Outer radius $= 14 + 7 = 21$ m.

Area of path $= \frac{22}{7} \times 21 \times 21 - \frac{22}{7} \times 14 \times 14 = 1386 - 616 = 770$ m^2.



(b) Cost $= 770 \times 40 =$ ₹ 30800.

(c) No. Outer radius $= 14 + 14 = 28$ m.

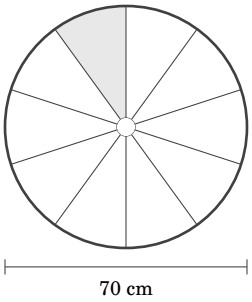
Area of path $= \frac{22}{7} \times 28 \times 28 - 616 = 2464 - 616 = 1848$ m^2.

Double the first path would be $2 \times 770 = 1540$ m^2. But 1848 m^2 is more than 1540 m^2.

The area of a circle depends on the square of its radius. So the outer part of a wider path adds more area than the inner part.

Section E (case-based question, 4 marks)

14. *Case study: the bicycle wheel.* Lakshmi rides her bicycle to school. Each wheel has a diameter of 70 cm. Each wheel has 10 spokes, equally spaced, that run from the centre to the rim.



(i) Find the distance the bicycle moves when the wheel turns once. [1]

(ii) Lakshmi's school is 1.1 km from her home. How many times does each wheel turn on her way to school? [1]

(iii) Find the area of the part of the wheel between two neighbouring spokes. [2]

OR

(iii) Find the perimeter of the part of the wheel between two neighbouring spokes. [2]

Answer.

(i) Radius = 35 cm. Distance in one turn = circumference = $2 \times \frac{22}{7} \times 35 = 220$ cm.

(ii) 1.1 km = 110000 cm. Number of turns = $110000 \div 220 = 500$.

(iii) The 10 spokes make 10 equal sectors, each of angle $360° \div 10 = 36°$.

Area of the wheel = $\frac{22}{7} \times 35 \times 35 = 3850$ cm^2. Area of one sector = $3850 \div 10 = 385$ cm^2.

OR

(iii) Arc of one sector = $220 \div 10 = 22$ cm. Perimeter = $22 + 35 + 35 = 92$ cm.

